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Team: 9

Github: https://github.com/TheodoreALSammour/k8s_challenge.git

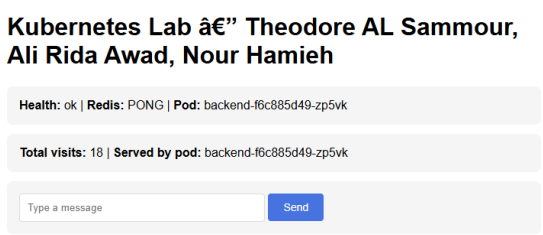
Screenshots:

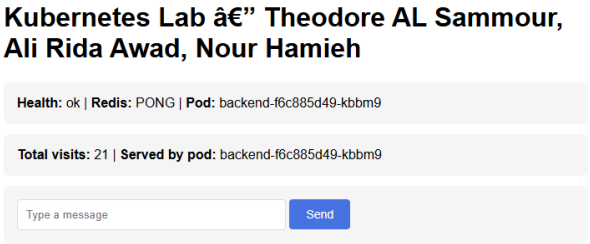
```
PS C:\Windows\system32> kubectl get nodes
NAME                                          STATUS   ROLES    AGE     VERSION
ip-192-168-11-3.eu-north-1.compute.internal   Ready    <none>   4m28s   v1.34.4-eks-f69f56f
ip-192-168-95-163.eu-north-1.compute.internal   Ready    <none>   4m27s   v1.34.4-eks-f69f56f
PS C:\Windows\system32> kubectl cluster-info
Kubernetes control plane is running at https://80D1E35D992AC36405C20AF9F58D0B86.gr7.eu-north-1.eks.amazonaws.com
CoreDNS is running at https://80D1E35D992AC36405C20AF9F58D0B86.gr7.eu-north-1.eks.amazonaws.com/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

```
C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>kubectl get pods -n k8s-lab
NAME                        READY   STATUS    RESTARTS   AGE
backend-f6c885d49-kbbm9     1/1     Running   0          2m23s
backend-f6c885d49-zp5vk     1/1     Running   0          2m10s
frontend-85c6c79965-24q4p   1/1     Running   0          11m
frontend-85c6c79965-bw8dl   1/1     Running   0          11m
redis-6c6fcd64b8-t5dlk      1/1     Running   0          2m32s

C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>
```

```
C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>kubectl get nodes -o wide
NAME                                          STATUS   ROLES    AGE   VERSION               INTERNAL-IP      EXTERNAL-IP    OS-IMAGE
  KERNEL-VERSION                  CONTAINER-RUNTIME
ip-192-168-11-3.eu-north-1.compute.internal     Ready    <none>   92m   v1.34.4-eks-f69f56f   192.168.11.3     51.20.89.155   Amazon Linux 2023.10.20260330
  6.12.77-99.140.amzn2023.x86_64   containerd://2.2.1+unknown
ip-192-168-95-163.eu-north-1.compute.internal   Ready    <none>   92m   v1.34.4-eks-f69f56f   192.168.95.163   13.60.214.9    Amazon Linux 2023.10.20260330
  6.12.77-99.140.amzn2023.x86_64   containerd://2.2.1+unknown
```

```
C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>kubectl get ingress -n k8s-lab
NAME          CLASS   HOSTS                ADDRESS                                                                   PORTS     AGE
app-ingress   nginx   16.192.1.159.nip.io   ac5dcb7ff58bb4787a2ac8f145124458-2c5176d5aa8e597f.elb.eu-north-1.amazonaws.com   80, 443   16m

C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>
```






```
C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>curl -k https://16.192.1.159.nip.io/api/health
{"status":"ok","redis":"PONG","pod":"backend-f6c885d49-zp5vk"}
C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>
```

```
C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>kubectl scale deployment frontend --replicas=4 -n k8s-lab
deployment.apps/frontend scaled

C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>kubectl get pods -n k8s-lab
NAME                        READY   STATUS    RESTARTS   AGE
backend-676bcc55dc-p7dzx    1/1     Running   0          79s
backend-676bcc55dc-rwpj8    1/1     Running   0          66s
frontend-85c6c79965-24q4p   1/1     Running   0          47m
frontend-85c6c79965-bw8dl   1/1     Running   0          47m
frontend-85c6c79965-mp8nf   1/1     Running   0          17s
frontend-85c6c79965-rx6zp   1/1     Running   0          17s
redis-6c6fcd64b8-t5dlk      1/1     Running   0          39m

C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>
```

```
C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>kubectl delete pod backend-676bcc55dc-p7dzx -n k8s-lab
pod "backend-676bcc55dc-p7dzx" deleted from k8s-lab namespace

C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>kubectl get pods -n k8s-lab -w
NAME                        READY   STATUS    RESTARTS   AGE
backend-676bcc55dc-rwpj8    1/1     Running   0          60m
backend-676bcc55dc-vd6j8    0/1     Running   0          10s
frontend-85c6c79965-24q4p   1/1     Running   0          106m
frontend-85c6c79965-bw8dl   1/1     Running   0          106m
frontend-85c6c79965-mp8nf   1/1     Running   0          59m
frontend-85c6c79965-rx6zp   1/1     Running   0          59m
redis-6c6fcd64b8-t5dlk      1/1     Running   0          98m
backend-676bcc55dc-vd6j8    1/1     Running   0          13s
```

```
C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>kubectl get hpa -n k8s-lab
NAME          REFERENCE            TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
backend-hpa   Deployment/backend   cpu: 3%/70%   2         6         2          3m18s

C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>
```

```
[ec2-user@ip-192-168-52-118 ~]$ kubectl get pods -n k8s-lab
NAME                        READY   STATUS    RESTARTS   AGE
backend-676bcc55dc-rwpj8    1/1     Running   0          77m
backend-676bcc55dc-vd6j8    1/1     Running   0          17m
frontend-85c6c79965-24q4p   1/1     Running   0          124m
frontend-85c6c79965-bw8dl   1/1     Running   0          124m
frontend-85c6c79965-mp8nf   1/1     Running   0          77m
frontend-85c6c79965-rx6zp   1/1     Running   0          77m
redis-6c6fcd64b8-t5dlk      1/1     Running   0          115m
[ec2-user@ip-192-168-52-118 ~]$ kubectl port-forward svc/backend-service 8080:3000 -n k8s-lab
Forwarding from 127.0.0.1:8080 -> 3000
Forwarding from [::1]:8080 -> 3000
Handling connection for 8080
Handling connection for 8080
```

```
[ec2-user@ip-192-168-52-118 ~]$ curl http://localhost:8080/api/health
{"status":"ok","redis":"PONG","pod":"backend-676bcc55dc-rwpj8"}[ec2-user@ip-192-168-52-118 ~]$ ^C
[ec2-user@ip-192-168-52-118 ~]$ curl http://localhost:8080/api/visits
{"visits":22,"pod":"backend-676bcc55dc-rwpj8"}[ec2-user@ip-192-168-52-118 ~]$ ^C
[ec2-user@ip-192-168-52-118 ~]$
```

```
C:\Users\theod\OneDrive\Desktop\AUB\Spring 25-26\EECE 503Q\K8s Challenge>eksctl delete cluster --name k8s-lab --region eu-north-1
2026-04-09 19:40:29 [ℹ]  deleting EKS cluster "k8s-lab"
2026-04-09 19:40:31 [ℹ]  will drain 0 unmanaged nodegroup(s) in cluster "k8s-lab"
2026-04-09 19:40:31 [ℹ]  starting parallel draining, max in-flight of 1
2026-04-09 19:40:32 [ℹ]  deleted 0 Fargate profile(s)
2026-04-09 19:40:35 [✓]  kubeconfig has been updated
2026-04-09 19:40:35 [ℹ]  cleaning up AWS load balancers created by Kubernetes objects of Kind Service, Ingress, or Gateway
2026-04-09 19:40:54 [ℹ]
2 sequential tasks: { delete nodegroup "workers", delete cluster control plane "k8s-lab" [async]
}
2026-04-09 19:40:55 [ℹ]  will delete stack "eksctl-k8s-lab-nodegroup-workers"
2026-04-09 19:40:55 [ℹ]  waiting for stack "eksctl-k8s-lab-nodegroup-workers" to get deleted
2026-04-09 19:40:55 [ℹ]  waiting for CloudFormation stack "eksctl-k8s-lab-nodegroup-workers"
2026-04-09 19:41:26 [ℹ]  waiting for CloudFormation stack "eksctl-k8s-lab-nodegroup-workers"
2026-04-09 19:42:04 [ℹ]  waiting for CloudFormation stack "eksctl-k8s-lab-nodegroup-workers"
2026-04-09 19:42:55 [ℹ]  waiting for CloudFormation stack "eksctl-k8s-lab-nodegroup-workers"
2026-04-09 19:43:50 [ℹ]  waiting for CloudFormation stack "eksctl-k8s-lab-nodegroup-workers"
2026-04-09 19:45:04 [ℹ]  waiting for CloudFormation stack "eksctl-k8s-lab-nodegroup-workers"
2026-04-09 19:46:58 [ℹ]  waiting for CloudFormation stack "eksctl-k8s-lab-nodegroup-workers"
2026-04-09 19:49:10 [ℹ]  waiting for CloudFormation stack "eksctl-k8s-lab-nodegroup-workers"
2026-04-09 19:49:10 [ℹ]  will delete stack "eksctl-k8s-lab-cluster"
2026-04-09 19:49:13 [✓]  all cluster resources were deleted
```